MILITARY SPECIFICATION SHEET

COATING, LOW REFLECTANCE ON GERMANIUM, 8.0 to 11.5, MICROMETERS, HIGH DURABILITY, SPECIAL REQUIREMENTS FOR

The complete requirements for procuring the low reflectance coating described herein shall consist of this document and the latest issue of Specification MIL-F-48616.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Spectral requirements exclusive of substrate absorptance for the low reflectance coating on germanium are given the following paragraphs:

1. The average transmittance through two coated surfaces shall be greater than 93% over the 8.0 to 11.5 micrometer region.

2. The minimum transmittance through two coated surfaces at any point between 8.0 and 11.5 micrometers shall not be less than 90%.

3. The average reflectance shall not be greater than 3.0% per surface over the 8.0 to 11.5 micrometer region.

4. The maximum reflectance at any point between 8.0 and 11.5 micrometers shall not be greater than 5.0% per surface.

Additional Durability Requirements:

In addition to the minimum acceptable durability requirements of MIL-F-48616 the coated element shall meet all of the optional durability requirements stated in 3.4.2.3.

NOTE:

Spectral resolution required to verify the above requirements shall be 0.1 micrometer or less. Scan speed and time constant shall be set such that scan time is two (2) minutes or greater for transmittance measurements and ten (10) minutes or greater for reflectance measurements.

Ordering Data:

Not applicable.

Preparing activity:
Army - AR
Project No. 6650-0091-2

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